

EC2 Network Connectivity Support Guide

Lab document - synthetic operational guidance for ingestion and retrieval testing.

Page 1 - Public IPv4 internet access

Incident EC2-NET-01

An EC2 instance has a public IPv4 address, but outbound HTTPS connections time out. Treat public addressing, routing, and packet filtering as separate checks.

Required conditions

For direct internet access over public IPv4, the VPC must have an internet gateway attached. The route table associated with the instance's subnet must contain a route for **0.0.0.0/0** whose target is that internet gateway. The instance must have a public IPv4 address or an Elastic IP address.

The instance security group must allow the required outbound traffic. Because security groups are stateful, response traffic for an allowed connection is automatically permitted. A custom network ACL is stateless, so its inbound and outbound rules must both allow the request and return path, including ephemeral ports when applicable.

Triage sequence

Step	Check	Expected state
1	Internet gateway	Attached to the same VPC
2	Subnet route	0.0.0.0/0 -> igw-id
3	Public address	Public IPv4 or Elastic IP assigned
4	Security group	Required egress allowed
5	Network ACL	Both directions and return ports allowed

A route to an internet gateway makes a subnet public, but it does not assign a public address to an instance and it does not override security controls.

Page 2 - Private subnet egress and reachability

Incident EC2-NET-02

An instance in a private subnet needs outbound software updates but must not accept unsolicited inbound internet connections.

Private IPv4 egress

Place a NAT gateway in a public subnet. The public subnet's route table sends 0.0.0.0/0 to an attached internet gateway. The private subnet's route table sends 0.0.0.0/0 to the NAT gateway. The private instance does not require a public IPv4 address.

Operational checks

Confirm that the private subnet is associated with the intended route table. Check the NAT gateway state and its public subnet. Then inspect security-group egress, network ACL rules, DNS resolution, and the destination's response. VPC Flow Logs can help distinguish accepted traffic from traffic rejected by a network ACL or security group.

Source references

AWS VPC subnet route tables:

<https://docs.aws.amazon.com/vpc/latest/userguide/subnet-route-tables.html>

AWS VPC route table options:

<https://docs.aws.amazon.com/vpc/latest/userguide/route-table-options.html>